

Alpha Code Analyzer: TCS AI Hackathon

Automated, multi-source code review and optimization.

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Team Alpha (TCS AI Hackathon) - 30/04/2026

Members:

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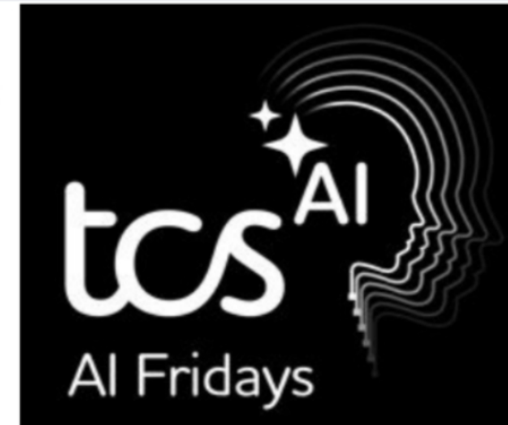
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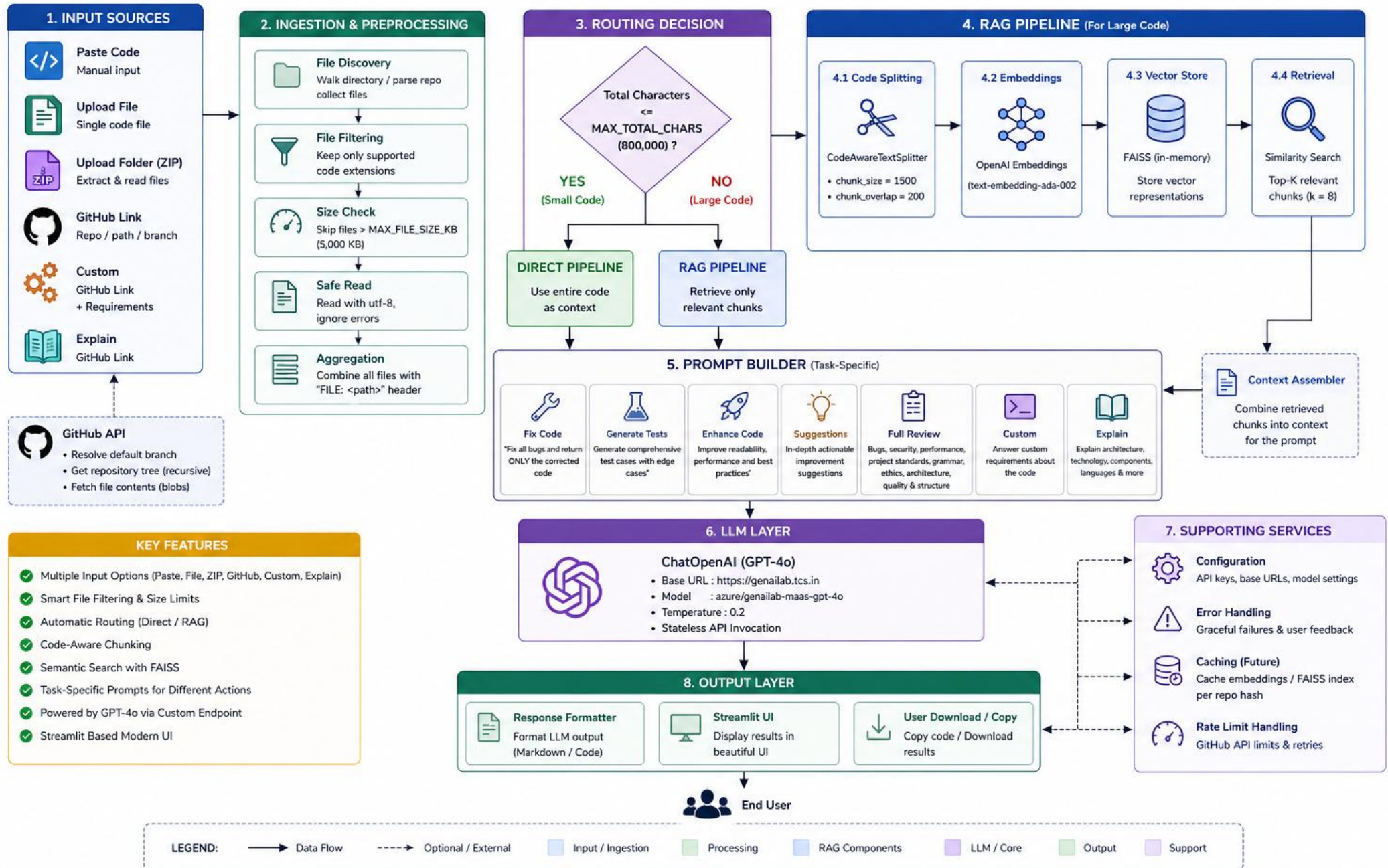
Gavin

TATA
CONSULTANCY
SERVICES



Alpha Code Analyzer – Architecture

RAG-Powered AI Code Review & Analysis Assistant



USE CASES – Real Impact Across TCS Domains

AI Code Assistant delivering tangible value in real-world TCS projects



1 Banking & Financial Services (BFSI)



Clients:



! Problem

- Legacy Java/Python codebases
- Manual code reviews take weeks
- High risk of bugs in transaction systems

💡 Our Solution

- Auto-analyze legacy code
- Suggest improvements
- Generate test cases for critical flows

📈 Impact

- Faster release cycles
- Reduced production bugs
- Cost savings in QA

2 IT Services / Large Codebase Maintenance



! Problem

- Multiple teams working on same repository
- Inconsistent code quality
- Code review bottlenecks

💡 Our Solution

- AI-powered pre-review assistant
- Standardized suggestions
- Auto-enhancement of code quality

📈 Impact

- Uniform coding standards
- Reduced review load
- Better developer onboarding

3 Manufacturing & Automation Systems



Clients:

TATA MOTORS

! Problem

- Embedded & control system code
- Bugs can cause real-world failures

💡 Our Solution

- Static analysis + bug detection
- Edge-case test generation

📈 Impact

- Reduced system failures
- Better reliability
- Safer deployments

4 Cloud Migration Projects



! Problem

- Old monolithic code → cloud-native
- Poor documentation & outdated patterns

💡 Our Solution

- Code enhancement suggestions
- Refactor recommendations
- Identify outdated patterns

📈 Impact

- Faster migration
- Better cloud-ready code
- Reduced rework

5 Developer Productivity Tool (Internal TCS)



! Problem

- New hires struggle with large codebases
- Time wasted understanding code

💡 Our Solution

- AI explains code
- Suggests improvements instantly
- Generates tests automatically

📈 Impact

- Faster onboarding
- Better understanding
- Reduced dependency on seniors

6 Secure Code Review (Cybersecurity)



! Problem

- Security vulnerabilities missed in manual reviews

💡 Our Solution

- Detect risky patterns
- Suggest secure alternatives

📈 Impact

- Improved security posture
- Fewer vulnerabilities
- Compliance-ready code



Business Impact



Up to
60%
reduction in manual
code review effort



Up to
40%
faster development
& release cycles



Significant improvement
in code quality, security
& compliance



One-line Impact

Our AI Assistant reduces manual code review effort by up to **60%** across large-scale enterprise projects delivered by TCS.

Bottlenecks in Manual Code Reviews

Engineering teams currently face a significant review tax, where manual code analysis consumes 30% of total development time, stalling productivity and preventing consistent quality across complex, growing codebases.

Missed Logic Errors

Human reviewers frequently fail to catch subtle logic errors or complex edge cases, particularly when operating under high-pressure, tight release cycle deadlines.

Complexity Limits

Teams struggle to maintain high standards of code review quality as the size of the repository grows and the underlying system architecture becomes more complex.

Technical Debt

Legacy codebases remain unoptimized because the manual effort required to refactor or rewrite problematic segments is prohibitively high for current engineering teams.

Context Switching

Developers lose their flow state while waiting hours or even days for feedback on simple syntax or logic improvements, drastically reducing overall output velocity.

Introducing the Alpha Code Analyzer Solution Overview

Direct LLM Logic

Utilizing advanced LLM capabilities to provide sophisticated reasoning on complex logic directly within the development environment.

Specialized Flow

Access dedicated workflows for fixing, testing, and optimizing code, streamlining development tasks with efficient single-click actions.



Universal Inputs

The system features flexible ingestion capabilities, seamlessly handling inputs ranging from single pasted functions to full GitHub repositories.

Fast Feedback

Accelerating the development process significantly by reducing the time-to-insight for code analysis from hours down to mere seconds.



Core Modules

The application is structured around five core actions, each utilizing specialized prompts to extract the best possible results from the LLM.



Detects syntax errors and logical bugs, returning **corrected** code.



Produces test cases focusing on **boundary** and edge conditions.



Analyzes bottlenecks and suggests optimizations for **efficiency**.



Provides improvements regarding **readability** and best practices.



Deep-dive analysis combining bugs, strategies, and **quality** scores.

Input Processing

We built a multi-modal analysis system designed to integrate directly into developer workflows via four distinct pathways.



Paste Code

High-speed text area input designed for immediate analysis of small, rapid code snippets.



File Upload

Direct ingestion of individual source files for highly targeted and specific code reviews.



ZIP Folder

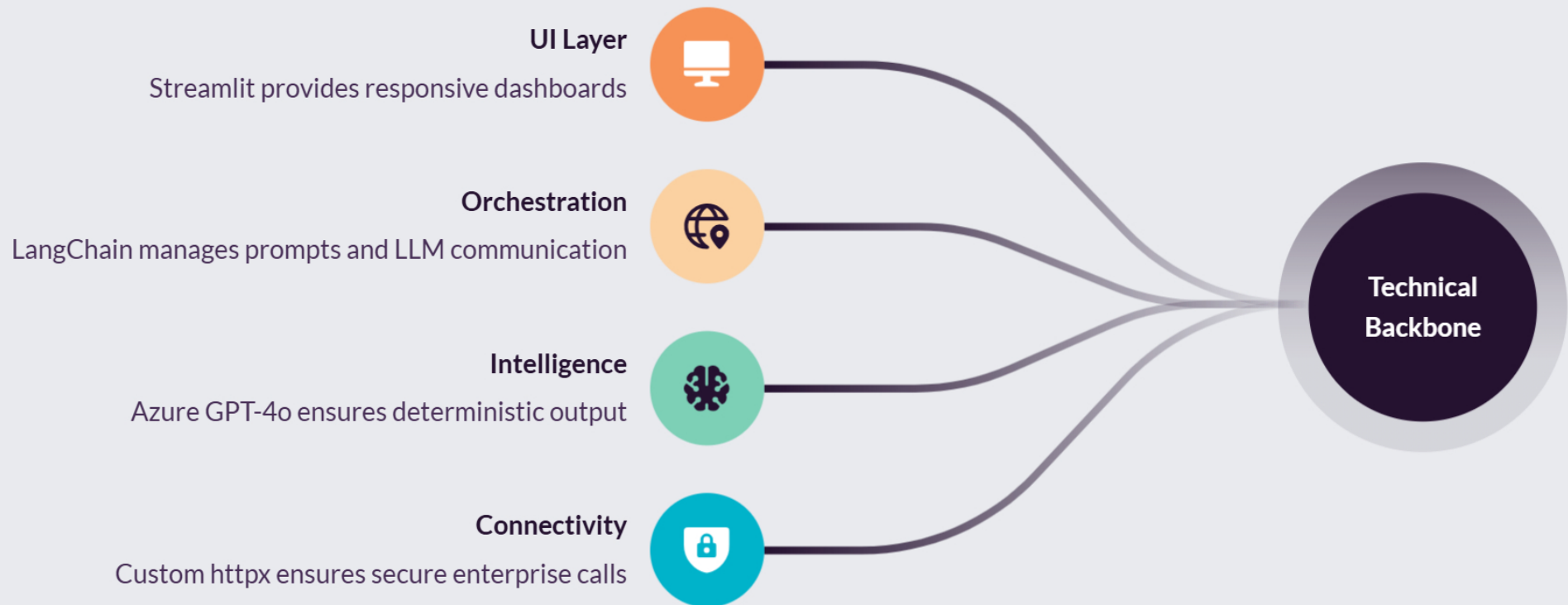
Utilizes tempfile extraction to traverse directories and locate supported extensions like .py and is



GitHub Link

Robust scraping engine that aggregates repositories into a unified buffer for full analysis.

Alpha Code Analyzer: Technical Stack and Core Infrastructure



GitHub Fetcher

Technical overview of automated repository ingestion and processing without requiring local clones.

01

URL Normalization

System automatically cleans user inputs, correctly handling .git suffixes and specific branch or blob file paths to ensure accurate repository targeting.

02

Branch Detection

The fetcher intelligently resolves the primary branch by sequentially checking standard defaults, specifically prioritizing main or master branch references.

03

Resource Optimization

Using TemporaryDirectory and ZipFile minimizes local disk usage by processing only relevant source files like .java, .cpp, and .ts formats efficiently.

04

Data Aggregation

Thousands of lines of code across nested directories are flattened into a single, LLM-readable context window to enable comprehensive cross-file analysis.

Dev Experience Shift

Alpha Code Analyzer improves speed, consistency, and coverage while reducing senior developer hours spent on manual reviews, elevating oversight capabilities.

Feature	Manual Review	Alpha Code Analyzer
Speed	30+ Minutes	~15 Seconds
Consistency	Subjective/Human Error	High/Deterministic
Scope	Limited focus	Full Repository
Unit Tests	Often skipped	Instant Generation
Cost	High Dev Hours	Low API Tokens

Future Engineering Excellence

By combining Streamlit UI with GPT-4o reasoning, the Alpha Code Analyzer reduces maintenance overhead, ensures high code quality, and accelerates development timelines to ship better code faster.

Open to questions!

Thank You
